

a one-period utility problem using most of the utility functions that we covered in chapter 2. All investors—whether they have short or long horizons—act like short-term investors in the i.i.d. world because returns are not predictable; the long-run investor faces a series of independent coin flips each period. The optimal strategy is to manage portfolio risk and return in each period, treating each period's asset allocation as a myopic investment problem. The optimal holding then is the myopic, short-run weight. The investor needs to rebalance back to this weight to avoid any one asset dominating in her portfolio for her given level of risk aversion.

If the optimal dynamic strategy is actually a myopic strategy, is the rebalancing strategy in Figure 4.4 different from a buy-and-hold strategy as shown in Figure 4.3? Absolutely. The dynamic problem is a series of one-period problems, and it involves *rebalancing* back to the same portfolio weight. The buy-and-hold problem involves doing nothing once the investor has bought at the beginning of the period. To rebalance back to the same weight, the investor has to trade each period.

Consider the simplest case of stocks and risk-free bonds. To maintain a fixed portfolio weight in stocks, an investor must invest *counter-cyclically*. If equity has done extremely well over the last period, equities now are above target and it is optimal to sell equity. Thus, the investor sells stocks when stocks have done well. Conversely, if equity loses money over the last period relative to other assets, equities have shrunk as a proportion of the total portfolio. The equity proportion is too low relative to optimal, and the investor buys equity. Thus, rebalancing buys assets that have gone down and sells assets that have gone up. This rebalancing is irrelevant to a myopic investor, because the myopic investor is not investing anymore after a single period. Rebalancing is the most basic and fundamental long-run investment strategy, and it is naturally counter-cyclical. An important consequence of optimal rebalancing is that long-run investors should actively divest from asset classes or even stocks that have done well and they should increase weights in asset classes or stocks that have low prices. Thus, rebalancing is a type of *value investing strategy* (see chapter 7): *long-run investors are at heart value investors*.

Rebalancing is optimal under i.i.d. returns, but it turns out to be advantageous when returns exhibit *mean reversion* or are predictable. If expected returns vary over time, prices are low because future expected returns are high—as our investor Daniel experienced during the 2008 financial crisis; prices of many risky assets plummeted, but their future expected returns from 2008 onwards were high. Rebalancing buys assets that have declined in price, which have high future expected returns. Conversely, rebalancing sells assets that have risen in price, which have low future expected returns.⁶

⁶I return to predictability in asset returns in chapter 8 and counter-cyclical factor investing in chapter 14.